

BIG DATA

Federico Valentino

f.valentino7@studenti.uniba.it

Beatrice Scavo

b.scavo@studenti.uniba.it

June 18, 2026

Contents

Introduction	3
Dataset.....	3
Dataset Preprocessing.....	3
Steps	4
First Step: Audience Filtering	4
Second Step: Format Filtering	4
Third Step: Keywords-based Filtering.....	6
Model and Fine Tuning.....	8
Fine Tuning Evaluation.....	9
Metrics	9
Results	9
Analysis and Technical Documentation of the RAG System	12
Objective of the Analysis.....	12
Configuration Mapping.....	12
System Architecture	14
Pipeline.....	14
Evaluation Metrics	14
Prompt	17
English Prompt.....	17
Italian Prompt	17
Modified Prompt.....	18
Questions	18
English Questions	18
Italian Questions.....	21
RAG Results	24
ROUGE-S	24
Average Metrics of Configurations	26
Limitations of ROUGE-S.....	29
LLM-as-a-Judge: Qualitative Evaluation Framework	32
BertScore	36
Limitations.....	39
Conclusions.....	39

Introduction

This project focuses on the preparation of a scientific dataset and the fine-tuning of a large language model for educational purposes.

The data used in this work was derived from *Cosmopedia*, a large-scale curated dataset designed for training language models on structured and pedagogical content.

The main objective of this project is to develop a model capable of generating clear, accurate, and educational explanations in scientific domains. To achieve this, the dataset was carefully filtered and reduced to include only relevant and samples about Physics, Chemistry and Biology. The model used for fine-tuning is *LLaMA 3.1 8B*, developed by *Meta*.

Dataset

The initial dataset originates from *Cosmopedia* and contains heterogeneous data in terms of:

- *subject domains*
- *audience levels*
- *textual formats*

Given this variability, it was necessary to restrict the dataset to a more controlled and relevant subset.

After filtering and selection, the dataset contains *42,327* samples, splitted up across three domains:

- *Chemistry*: 19,313
- *Physics*: 22,491
- *Biology*: 523

Additionally, during the filtering process, a new column called *subject_type* was introduced. This field explicitly identifies the scientific domain of each example enabling:

- better dataset organization
- easier analysis and evaluation

Dataset Preprocessing

The preprocessing phase was essential to ensure the quality, consistency, and relevance of the dataset. Starting from the original *Cosmopedia* dataset, this process involved multiple filtering steps based on the `<audience>` and `<format>` fields, followed by an additional keyword-based filtering stage to extract domain-specific content, as well as dataset size tracking at each stage.

Steps

First Step: Audience Filtering

The dataset was first filtered based on the <audience> field.

Audience Field Values

- *young_children* → very young children (early childhood / pre-school)
- *grade_school_students* → primary school
- *middle_school_students* → middle school
- *high_school_students* → high school
- *college_students* → university level
- *researchers* → postgraduate / research level
- *general* → general audience (not tied to a specific educational level)

Removed <audience> values:

- *college_students*
- *researchers*
- *general*

Files completely removed as not relevant:

- khanacademy
- stories
- wikihow

Dataset Size After <audience> Preprocessing

- auto_math_text: from 1,949,895 to 979,915
- openstax: from 126,332 to 95,306
- stanford: from 1,020,024 to 511,252

Total: 1,586,473

Second Step: Format Filtering

A second filtering stage was applied based on the <format> field.

Format Field Values

1. **story_reddit**

A narrative in the style of a Reddit post: informal, first-person perspective, often including an initial problem, development of events, and a conclusion. Uses colloquial and realistic language, sometimes with dialogue or personal commentary.

2. **story_life_lessons**

A narrative that conveys an explicit life lesson. Simple structure: event → difficulty → reflection → moral (e.g., perseverance, empathy, responsibility).

3. **story_children**

A children's story: simple language, short sentences, clear concepts, often involving symbolic characters or talking animals. Educational but accessible and reassuring.

4. **story_morality**

A story focused on ethical dilemmas or moral values. Often more serious in tone, aiming to encourage reflection on right and wrong, responsibility, and consequences of actions.

5. **story_forums**

A narrative in the style of an online forum: semi-informal tone, description of a personal experience with an implicit or explicit request for opinions. Typically structured as context → problem → question to the community.

6. **textbook_academic_tone**

A formal, objective, and structured educational text. Uses technical language, precise definitions, and systematically organized concepts. No personal or narrative elements.

7. **blogpost**

An informative article with an accessible yet competent tone. Fluid structure, practical examples, and possible use of rhetorical questions or author opinions to engage the reader.

8. **wikihow**

A step-by-step practical guide oriented toward action. Procedural structure with numbered steps, often including tips and warnings.

9. **textbook_narrative_tone**

An educational text that maintains conceptual rigor while using narrative examples or case studies to explain concepts. Less rigid than purely academic tone but still structured and formal.

Removed <format> Values

- story_life_lessons
- story_children
- story_morality

Dataset Size After <format> Preprocessing

- **auto_math_text**: from 979,915 to 979,915
- **openstax**: from 95,306 to 95,306
- **stanford**: from 511,252 to 511,252

Total: 1,586,473

From the preprocessing results, it can be observed that the <audience> filtering step had a significant impact on the dataset size, removing a large portion.

In contrast, the subsequent <format> filtering step did not lead to any further reduction in the dataset. A key factor in this behaviour is the removal of the *general* audience category. Since this category typically includes a wide variety of formats—including more informal or narrative styles—its exclusion likely contributed to an implicit alignment of the dataset toward more structured and educational formats.

Third Step: Keywords-based Filtering

Due to computational constraints related to the number of samples used for fine-tuning, only the files belonging to *auto_math_text* were selected and a keyword-based filtering script was developed to extract samples related to Physics, Chemistry, and Biology to ensure domain-specific relevance.

Filtering idea

The used script searches for keywords across <text> field. It uses predefined keyword lists for each scientific domain and matches them using regular expressions to ensure precise detection. A row is kept only if it contains at least a minimum number of keyword matches, ensuring higher relevance and reducing noise.

The set of keywords was obtained by combining two approaches:

1. Including terms based on our own knowledge of the domain of interest;
2. **AI Detector**: attempting to identify, within the document collection, which words were used for the domain through an LLM. A small number of examples was therefore provided as input prompts to ChatGPT, which then identified the words in the texts that it considered most discriminative for that domain. If, for that set of examples, it did not recognize any instance belonging to the domain, a different set of examples was selected and provided again as input.

Note: The limitation on the number of examples submitted to ChatGPT is due to the prompt token constraints, as each dataset example contains a large number of words.

After applying the keyword-based filtering script, the resulting files were further manually reviewed and supported by an LLM-based (ChatGPT) validation step to perform an additional quality check. This step helped identify and remove any remaining irrelevant or misclassified samples, improving the overall consistency and reliability of the dataset.

Used Keywords

PHYSICS_KEYWORD =

"physics", "physical quantity", "measurement", "unit", "si unit", "scalar", "vector", "magnitude", "direction", "motion", "kinematics", "dynamics", "force", "newton's laws", "mass", "acceleration", "velocity", "speed", "momentum", "impulse", "inertia", "friction", "gravity", "gravitational force", "work", "energy", "kinetic energy", "potential energy", "power", "conservation of energy", "electricity", "electric current", "current", "voltage", "electric potential", "resistance", "ohm's law", "kirchhoff's laws", "circuit", "series circuit", "parallel circuit", "electromagnetism", "electric field", "magnetic field", "charge", "coulomb", "coulomb's law", "electric flux", "magnetic flux", "faraday's law", "electromagnetic induction", "lorentz force", "capacitor", "inductor", "resistor", "battery", "power supply", "wave", "frequency", "wavelength", "amplitude", "wave speed", "oscillation", "period", "optics", "light", "reflection", "refraction", "diffraction", "interference", "lens", "mirror", "thermodynamics", "heat", "temperature", "entropy", "internal energy", "heat transfer", "conduction", "convection", "radiation", "quantum", "quantum mechanics", "photon", "energy level", "wave-particle duality", "uncertainty principle", "relativity", "time dilation", "mass-energy equivalence"

CHEMISTRY_KEYWORDS =

"chemistry", "matter", "substance", "pure substance", "mixture", "element", "compound", "atom", "molecule", "ion", "cation", "anion", "periodic table", "electron", "proton", "neutron", "nucleus", "orbital", "electron configuration", "valence electron", "energy level", "isotope", "atomic number", "chemical bond", "ionic bond", "covalent bond", "metallic bond", "electronegativity", "lewis structure", "octet rule", "intermolecular forces", "stoichiometry", "mole", "molar mass", "avogadro's number", "chemical equation", "reactant", "product", "yield", "reaction", "reaction rate", "combustion", "redox", "acid-base", "synthesis", "decomposition", "catalyst", "solid", "liquid", "gas", "phase change", "enthalpy", "endothermic", "exothermic", "calorimetry", "solution", "solute", "solvent", "molarity", "dilution", "solubility", "acid", "base", "ph", "buffer", "titration", "neutralization", "equilibrium", "equilibrium constant", "le chatelier's principle", "oxidation", "reduction", "electrode", "anode", "cathode", "electrolysis", "hydrocarbon", "alkane", "alkene", "alcohol", "ketone", "polymer", "radioactivity", "decay", "half-life", "fission", "fusion"

BIOLOGY_KEYWORDS =

"biology", "organism", "life", "photosynthesis", "respiration", "metabolism", "diffusion", "osmosis", "homeostasis", "cell", "cell membrane", "nucleus", "mitochondria", "ribosome", "cytoplasm", "organelle", "dna", "rna", "gene", "chromosome", "mutation", "protein synthesis", "protein", "lipid", "carbohydrate", "enzyme", "atp", "tissue", "organ", "nervous system", "circulatory system", "immune system", "hormone", "feedback mechanism", "bacteria", "virus", "microorganism", "microscope", "experiment", "specimen".

Despite the multiple filtering steps and additional validation, it is important to note that some level of noise in the dataset is inevitable. While significant effort was made to ensure domain relevance, automatic and semi-automatic filtering methods cannot guarantee perfect accuracy.

Furthermore, although the dataset was designed to be as balanced as possible across the three domains, a strong imbalance remains: Biology related samples are less represented compared to Physics and Chemistry, due to the lower availability of highly relevant data after the filtering process.

Model and Fine Tuning

The base model adopted in this work is LLaMA 3.1 8B, a transformer-based causal language model optimized for next-token prediction tasks. The model was adapted to the target scientific domains through Supervised Fine-Tuning (SFT) on the curated dataset.

To reduce memory footprint and computational cost, fine-tuning was performed using LoRA (Low-Rank Adaptation), which injects trainable low-rank matrices into the attention projection layers while keeping the base model weights frozen. The projection layers interested are $\{q_{proj}, k_{proj}, v_{proj}, o_{proj}\}$.

Fine-tuning is performed using QLoRA, which combines LoRA (Low-Rank Adaptation) with 4-bit quantization to reduce memory usage while keeping most of the model weights frozen.

Training is performed using the TRL SFTTrainer, with the following configuration: batch size 2, gradient accumulation steps 8, maximum sequence length 1024, learning rate 2×10^{-4} , and 5 training epochs. The optimizer used is paged_adamw_8bit in GPU.

The training objective is standard Causal Language Modeling, where the model minimizes the Cross-Entropy loss between predicted tokens and the ground-truth next token:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(x_i | x_{<i})$$

where x_i is the target token and $x_{<i}$ represents the preceding context.

This formulation encourages the model to assign high probability to the correct next token at each position in the sequence.

The entire training loop, including forward pass, loss computation, gradient updates, logging, and checkpointing, is managed by the SFTTrainer. At the end of training, both the fine-tuned model and tokenizer are saved for downstream usage.

Fine Tuning Evaluation

Metrics

- **Mean token accuracy**

Average percentage of tokens correctly predicted by the model.

$$\frac{N_{\text{correct}}}{N_{\text{total}}}$$

- **Entropy**

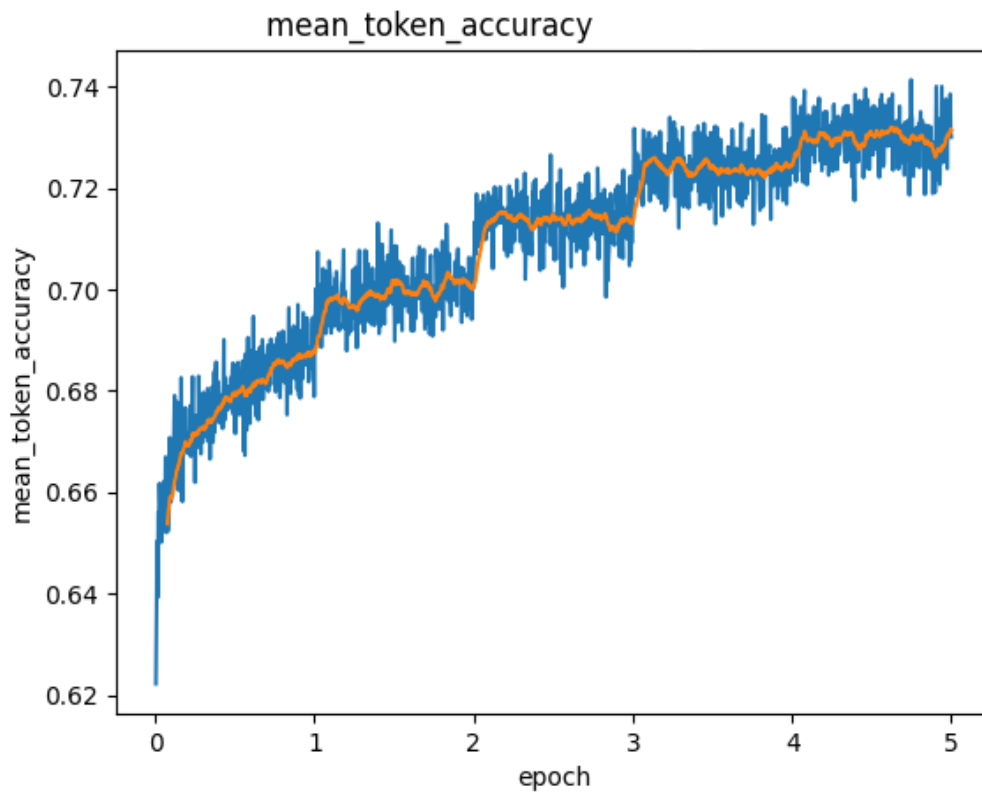
Quantifies the uncertainty of a probability distribution.

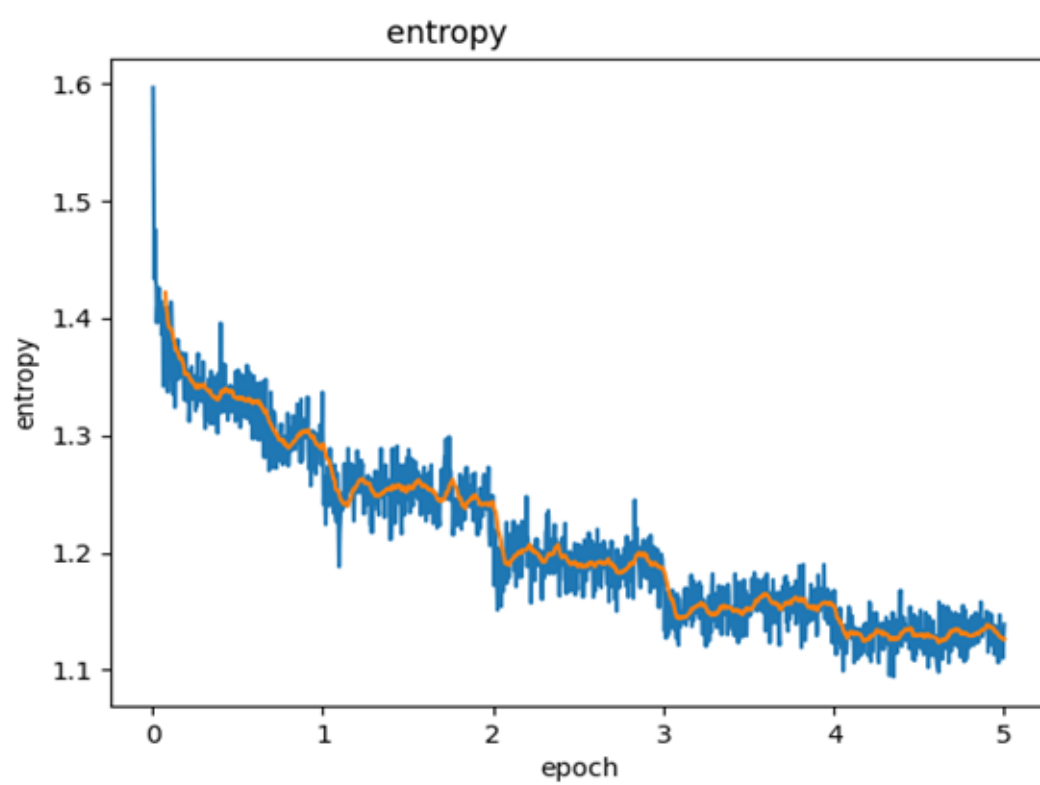
$$H(Q) = - \sum_{k=1}^K q_k \log(q_k)$$

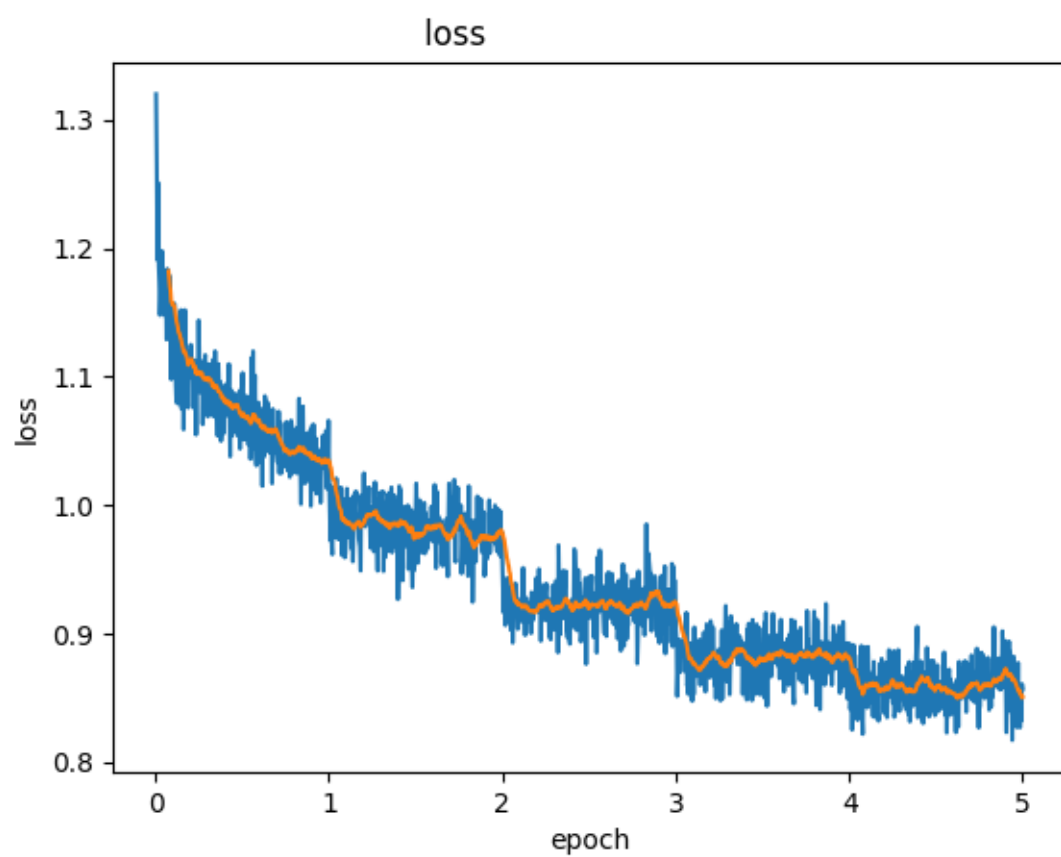
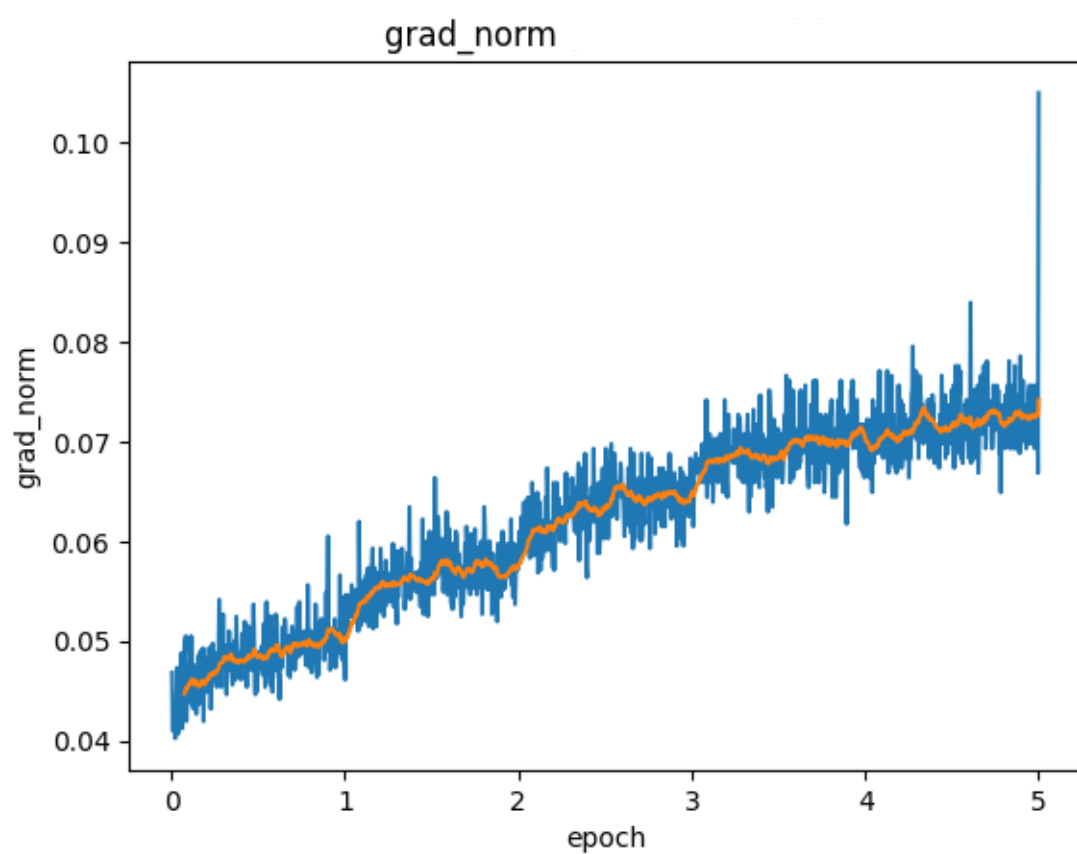
- **GradNorm**

Measures the magnitude of the parameter gradients and indicates how large the model updates are during training

Results







Analysis and Technical Documentation of the RAG System

Objective of the Analysis

This section analyzes the performance of different configurations of a Retrieval-Augmented Generation (RAG) system.

The experimental configurations were defined by combining the following factors:

- language of the retrieval dataset (Italian or English)
- language of the queries (Italian or English)
- decoding strategy (DoSample = TRUE or DoSample = FALSE)
- model type (base or fine-tuned)

In particular, the following configurations were considered:

- Italian RAG dataset translated into English (via Deep Translator), with English queries
- Italian RAG dataset (original), with Italian queries
- English RAG dataset, with English queries

For each configuration, the following variables were controlled:

- decoding strategy
- model type (base / fine-tuned)

The analysis was conducted on 12 distinct CSV files, each representing a specific system configuration.

Performance was evaluated using the ROUGE-S metric, which measures lexical overlap between the generated answer and the reference.

Configuration Mapping

Each CSV file corresponds to a specific RAG configuration defined by:

- dataset language
- query language
- decoding strategy
- model type

File	Dataset Language	Question Language	Answer Language	DoSample	Model
rag_results_rouge_s DoSampleFALSE_Modello Base.csv	English (translated)	English	English	FALSE	Base
rag_results_rouge_s DoSampleTRUE_Modello Base.csv	English (translated)	English	English	TRUE	Base
rag_results_rouge_s DoSampleFALSE.csv	English (translated)	English	English	FALSE	Fine-tuned
rag_results_rouge_s DoSampleTRUE.csv	English (translated)	English	English	TRUE	Fine-tuned
rag_results_rouge_s_IT DoSampleFALSE_Modello Base.csv	Italian	Italian	Italian	FALSE	Base
rag_results_rouge_s_IT DoSampleTRUE_Modello Base.csv	Italian	Italian	Italian	TRUE	Base
rag_results_rouge_s_IT DoSampleFALSE.csv	Italian	Italian	Italian	FALSE	Fine-tuned
rag_results_rouge_s_IT DoSampleTRUE.csv	Italian	Italian	Italian	TRUE	Fine-tuned
ragEN_results_rouge_s DoSampleFALSE_Modello Base.csv	English	English	English	FALSE	Base
ragEN_results_rouge_s DoSampleTRUE_Modello Base.csv	English	English	English	TRUE	Base
ragEN_results_rouge_s DoSampleFALSE.csv	English	English	English	FALSE	Fine-tuned
ragEN_results_rouge_s DoSampleTRUE.csv	English	English	English	TRUE	Fine-tuned

Methodological Note

The translated Italian RAG dataset was obtained using Deep Translator in order to align the language of the context with that of the queries in certain configurations.

System Architecture

The system is based on a Retrieval-Augmented Generation framework composed of:

- a retrieval module based on text embeddings
- a generation module based on a large language model
- an automatic evaluation module

Pipeline

The system follows a three-stage pipeline:

Retrieval

- documents are split into chunks
- chunks are encoded using embedding models
- the most relevant embedding chunks are retrieved based on similarity with the embedding query with cosine similarity measure

Generation

- retrieved chunks are inserted into a structured prompt
- the model generates an answer constrained by the provided context

Evaluation

- the generated answer is compared to the retrieved context using:
 - ROUGE-S
 - LLM-as-a-Judge (Qualitative Evaluation)
 - BERTScore (Semantic Evaluation)

Evaluation Metrics

ROUGE-S (Skip-Bigram)

The main evaluation metric used is ROUGE-S, which measures similarity between two texts based on skip-bigrams (ordered word pairs).

Definition:

Given a sequence of words, ROUGE-S considers all ordered pairs:

$$(w_i, w_j), i < j$$

The system computes:

- **Precision**

$$\frac{\text{overlap}}{\text{skip-bigram of generated text}}$$

- **Recall**

$$\frac{\text{overlap}}{\text{skip – bigram of the reference text}}$$

- **F1-score**

$$\frac{2PR}{P + R}$$

ROUGE-S captures:

- lexical overlap
- structural similarity

However, it does not measure:

- semantic correctness
- quality of explanation

In this system, the reference is the retrieved context, not a manually curated ground-truth answer.

Qualitative Evaluation (LLM-as-a-Judge)

To complement the limitations of lexical metrics, a qualitative evaluation was performed using an LLM-based judge.

Each generated answer was classified into one of the following categories:

- Coherent
- Partially coherent
- Not coherent

This evaluation framework allows for a more comprehensive assessment of model performance by capturing:

- semantic correctness
- completeness of the response
- practical usefulness of the generated answer

Unlike automatic metrics, this approach provides a higher-level interpretation of answer quality, making it particularly suitable for evaluating educational and explanatory outputs.

BERTScore (Semantic Evaluation)

To address the limitations of ROUGE-S, an additional evaluation metric based on semantic similarity, BERTScore, was introduced.

BERTScore computes similarity between the generated answer and the reference text by leveraging contextual embeddings derived from pre-trained transformer models such as BERT. Instead of relying on exact lexical overlap, it evaluates how semantically close two texts are by comparing token representations in a high-dimensional embedding space.

In this work, the model used for semantic evaluation is:

- XLM-RoBERTa (xlm-roberta-large)

Differently from retrieval embeddings used in the RAG pipeline, BERTScore generates contextual token-level embeddings independently for:

- the generated answer (candidate text)
- the reference text

The metric is computed by aligning each token in the generated text with the most semantically similar token in the reference text using cosine similarity between embeddings.

The cosine similarity between two token embeddings is defined as:

$$\text{sim}(x_i, y_j) = \frac{x_i^\top y_j}{\|x_i\| \|y_j\|}$$

Based on these token alignments, precision, recall, and F1-score are computed.

- **Precision:** measures how much of the generated text is semantically supported by the reference

$$P = \frac{1}{|x|} \sum_{i=1}^{|x|} \max_j \text{sim}(x_i, y_j)$$

- **Recall:** measures how much of the reference content is captured by the generated text

$$R = \frac{1}{|y|} \sum_{j=1}^{|y|} \max_i \text{sim}(x_i, y_j)$$

- **F1-score:** harmonic mean of precision and recall

$$F1 = 2 \cdot \frac{P \cdot R}{P + R}$$

Unlike ROUGE-S, BERTScore captures:

- semantic similarity
- paraphrasing robustness
- contextual meaning

This makes it particularly suitable for evaluating generative models that produce abstractive and rephrased responses.

Prompt

English Prompt

You are a scientific assistant.

STRICT RULES:

- Use ONLY the information from the provided context.
- DO NOT use prior knowledge.
- DO NOT add information that is not explicitly stated.
- If any step is missing, say: Not available in the provided context.

CONTEXT:

{context}

QUESTION:

{question}

ANSWER:

Italian Prompt

Sei un assistente scientifico.

REGOLE RIGIDE:

- Usa SOLO le informazioni presenti nel contesto fornito.
- NON usare conoscenze pregresse.
- NON aggiungere informazioni che non sono esplicitamente presenti.
- Rispondi SEMPRE in italiano.
- Se manca qualche informazione, scrivi: Non disponibile nel contesto fornito.

CONTESTO:

{context}

DOMANDA:

{question}

RISPOSTA:

Modified Prompt

Removal of the following strict rules:

- Use ONLY the information from the provided context.
- DO NOT use prior knowledge.
- DO NOT add information that is not explicitly stated.
- If any step is missing, say: Not available in the provided context.

for English;

- Usa SOLO le informazioni presenti nel contesto fornito.
- NON usare conoscenze pregresse.
- NON aggiungere informazioni che non sono esplicitamente presenti.
- Se manca qualche informazione, scrivi: Non disponibile nel contesto fornito.

for Italian;

Questions

English Questions

```
questions = [
```

```
# Esperimento_1_Scienze_della_vita
```

```
"Can you extract a list of the steps required to carry out the Lavoisier's law experiment and verify the principle of conservation of mass?",
```

```
"What is the objective of the Lavoisier's law experiment regarding the conservation of mass in the reaction between zinc and hydrochloric acid?",
```

```
"I carried out the experiment leaving the flask open during the reaction between zinc crystals and hydrochloric acid: is this correct? If not, what should I correct to perform the experiment properly?",
```

```
"How can I correctly complete the experiment using a flask, zinc, and hydrochloric acid to verify that the total mass remains constant according to Lavoisier's law?",
```

```
# Esperimento_Elettricità
```

```
"Can you extract a list of the steps required to carry out the digital multimeter experiment and correctly measure resistance, voltage, and current?",
```

```
"What is the objective of the digital multimeter experiment regarding the measurement of resistance, voltage (potential difference), and current intensity?",
```

```
"I carried out the experiment by measuring the resistance of a component while it was still powered: is this correct? If not, what should I correct to perform the experiment properly?",
```

"How can I correctly complete the experiment using a digital multimeter, probes, and a battery to measure voltage, current, and resistance accurately?",

Esperimento_biologia_modificato

"Can you extract a list of the steps required to carry out the plant cell observation experiment and compare the structure of onion and orchid cells under a microscope?",

"What is the objective of the plant cell experiment regarding the comparison between onion cells and Orchid Phalaenopsis cells observed under a microscope?",

"I carried out the experiment placing the cover slip without paying attention to air bubbles during the preparation of the onion epidermis sample: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using a microscope, slides, onion epidermis, and methylene blue to observe and compare plant cells accurately?",

Esperimento_elettromagnetismo_modificato

"Can you extract a list of the steps required to carry out the magnet experiment and observe the interaction between magnetic poles?",

"What is the objective of the magnet experiment regarding the interaction between two magnets and their poles?",

"I carried out the experiment by bringing the magnets close together without changing their orientation to test different poles: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using two magnets to observe the attraction and repulsion between opposite and like poles?",

Esperimento_fototropismo_modificato

"Can you extract a list of the steps required to carry out the phototropism experiment and observe how plants grow in response to light direction?",

"What is the objective of the phototropism experiment regarding the growth of plants under different light conditions and the distinction between positive and negative phototropism?",

"I carried out the experiment by exposing all plants to uniform light instead of directing light from one side in the experimental setup: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using seeds, soil, boxes, and a controlled light source to observe and measure phototropism in plants accurately?",

Esperimento_meccanica_modificato

"Can you extract a list of the steps required to carry out the caliper experiment and measure the dimensions of a solid object accurately?",

"What is the objective of the caliper experiment regarding the measurement of width, length, and depth of an object using a vernier caliper?",

"I carried out the experiment by reading only the main scale of the caliper without considering the vernier scale: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using a vernier caliper and a solid object to measure dimensions with precision, including reading both the main scale and the vernier scale?",

Estrazione_dna_vegetale_sintetico

"Can you extract a list of the steps required to carry out the plant DNA extraction experiment and obtain visible DNA from plant cells?",

"What is the objective of the DNA extraction experiment regarding the isolation and visualization of DNA from plant cells using common laboratory materials?",

"I carried out the experiment by using alcohol at room temperature instead of cold alcohol during the DNA precipitation step: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using plant material, detergent, salt, and cold alcohol to extract and observe DNA accurately?",

Manuale_energia_alternativa_modificato

"Can you extract a list of the steps required to carry out the internal combustion engine experiment and observe how fuel ignition occurs inside a cylinder?",

"What is the objective of the internal combustion engine experiment regarding the functioning of fuel combustion and ignition inside a confined system?",

"I carried out the experiment by increasing the amount of alcohol sprayed inside the cylinder without considering the available oxygen: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using a cylinder, alcohol, and a spark source to observe and understand the combustion process accurately?",

Manuale_la_termodinamica_modificato

"Can you extract a list of the steps required to carry out the distillation experiment and separate alcohol from a water-alcohol mixture?",

"What is the objective of the distillation experiment regarding the separation of substances based on their different boiling points?",

"I carried out the experiment without cooling the U-shaped tube during the distillation process: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using a distillation setup, heat source, and cooling system to separate and collect alcohol from a mixture accurately?",

Saggio-fiamma_libro_zanichelli

"Can you extract a list of the steps required to carry out the flame test experiment and identify different metal ions based on flame color?",

"What is the objective of the flame test experiment regarding the identification of metals through their characteristic emission of light?",

"I carried out the experiment without cleaning the wire loop between testing different metal salts: is this correct? If not, what should I correct to perform the experiment properly?",

"How can I correctly complete the experiment using a flame source, metal salts, and a wire loop to accurately observe and distinguish the characteristic colors of different elements?",

]

Italian Questions

questions = [

Esperimento_1_Scienze_della_vita

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento sulla legge di Lavoisier e verificare il principio di conservazione della massa?",

"Qual è l'obiettivo dell'esperimento sulla legge di Lavoisier riguardo alla conservazione della massa nella reazione tra zinco e acido cloridrico?",

"Ho svolto l'esperimento lasciando il matraccio aperto durante la reazione tra cristalli di zinco e acido cloridrico: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un matraccio, zinco e acido cloridrico per verificare che la massa totale rimanga costante secondo la legge di Lavoisier?",

Esperimento_Elettricità

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento con il multimetro digitale e misurare correttamente resistenza, tensione e corrente?",

"Qual è l'obiettivo dell'esperimento con il multimetro digitale riguardo alla misura di resistenza, tensione (differenza di potenziale) e intensità di corrente?",

"Ho svolto l'esperimento misurando la resistenza di un componente mentre era ancora alimentato: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un multimetro digitale, i puntali e una batteria per misurare con precisione tensione, corrente e resistenza?",

Esperimento_biologia_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento di osservazione delle cellule vegetali e confrontare al microscopio la struttura delle cellule di cipolla e orchidea?",

"Qual è l'obiettivo dell'esperimento sulle cellule vegetali riguardo al confronto tra cellule di cipolla e cellule di Orchidea Phalaenopsis osservate al microscopio?",

"Ho svolto l'esperimento posizionando il coprioggetto senza fare attenzione alle bolle d'aria durante la preparazione del campione di epidermide di cipolla: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un microscopio, vetrini, epidermide di cipolla e blu di metilene per osservare e confrontare accuratamente le cellule vegetali?",

Esperimento_elettromagnetismo_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento sui magneti e osservare l'interazione tra i poli magnetici?",

"Qual è l'obiettivo dell'esperimento sui magneti riguardo all'interazione tra due magneti e i loro poli?",

"Ho svolto l'esperimento avvicinando i magneti senza cambiarne l'orientamento per testare poli diversi: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando due magneti per osservare l'attrazione e la repulsione tra poli uguali e opposti?",

Esperimento_fototropismo_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento sul fototropismo e osservare come le piante crescono in risposta alla direzione della luce?",

"Qual è l'obiettivo dell'esperimento sul fototropismo riguardo alla crescita delle piante in diverse condizioni di luce e alla distinzione tra fototropismo positivo e negativo?",

"Ho svolto l'esperimento esponendo tutte le piante a una luce uniforme invece di dirigere la luce da un solo lato nella configurazione sperimentale: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando semi, terreno, scatole e una sorgente luminosa controllata per osservare e misurare accuratamente il fototropismo nelle piante?",

Esperimento_meccanica_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento con il calibro e misurare con precisione le dimensioni di un oggetto solido?",

"Qual è l'obiettivo dell'esperimento con il calibro riguardo alla misura di larghezza, lunghezza e profondità di un oggetto usando un calibro a corsoio?",

"Ho svolto l'esperimento leggendo solo la scala principale del calibro senza considerare il nonio: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un calibro a corsoio e un oggetto solido per misurare le dimensioni con precisione, leggendo sia la scala principale sia il nonio?",

Estrazione_dna_vegetale_sintetico

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento di estrazione del DNA vegetale e ottenere DNA visibile dalle cellule vegetali?",

"Qual è l'obiettivo dell'esperimento di estrazione del DNA riguardo all'isolamento e alla visualizzazione del DNA da cellule vegetali usando materiali comuni di laboratorio?",

"Ho svolto l'esperimento usando alcol a temperatura ambiente invece di alcol freddo durante la fase di precipitazione del DNA: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando materiale vegetale, detergente, sale e alcol freddo per estrarre e osservare accuratamente il DNA?",

Manuale_energia_alternativa_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento sul motore a combustione interna e osservare come avviene l'accensione del carburante all'interno di un cilindro?",

"Qual è l'obiettivo dell'esperimento sul motore a combustione interna riguardo al funzionamento della combustione del carburante e dell'accensione all'interno di un sistema confinato?",

"Ho svolto l'esperimento aumentando la quantità di alcol spruzzata nel cilindro senza considerare l'ossigeno disponibile: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un cilindro, alcol e una sorgente di scintilla per osservare e comprendere accuratamente il processo di combustione?",

Manuale_la_termodinamica_modificato

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento di distillazione e separare l'alcol da una miscela acqua-alcol?",

"Qual è l'obiettivo dell'esperimento di distillazione riguardo alla separazione di sostanze in base ai loro diversi punti di ebollizione?",

"Ho svolto l'esperimento senza raffreddare il tubo a U durante il processo di distillazione: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando un apparato di distillazione, una sorgente di calore e un sistema di raffreddamento per separare e raccogliere accuratamente l'alcol da una miscela?",

Saggio-fiamma_libro_zanichelli

"Puoi estrarre un elenco dei passaggi necessari per svolgere l'esperimento del saggio alla fiamma e identificare diversi ioni metallici in base al colore della fiamma?",

"Qual è l'obiettivo dell'esperimento del saggio alla fiamma riguardo all'identificazione dei metalli attraverso la loro caratteristica emissione di luce?",

"Ho svolto l'esperimento senza pulire l'ansa di filo tra la prova di diversi sali metallici: è corretto? Se no, cosa dovrei correggere per eseguire bene l'esperimento?",

"Come posso completare correttamente l'esperimento usando una sorgente di fiamma, sali metallici e un'ansa di filo per osservare e distinguere accuratamente i colori caratteristici dei diversi elementi?",

]

RAG Results

ROUGE-S

The above-mentioned metrics (Precision, Recall, F1-score) were computed for each experimental configuration:

- Italian dataset: DoSample=True/False, Base Model/Fine-Tuned Model
- English dataset (previously translated): DoSample=True/False, Base Model / Fine-Tuned Model
- English dataset (translated with DeepTranslator): DoSample=True/False, Base Model/ Fine-Tuned Model

resulting in a total of 18 graphs are available in “Plots” folder.

How to read the plots:

Each graph shows the represented metric on the y-axis and all 40 questions on the x-axis, grouped by the 10 documents and labeled as

<question_number>-<document_number>

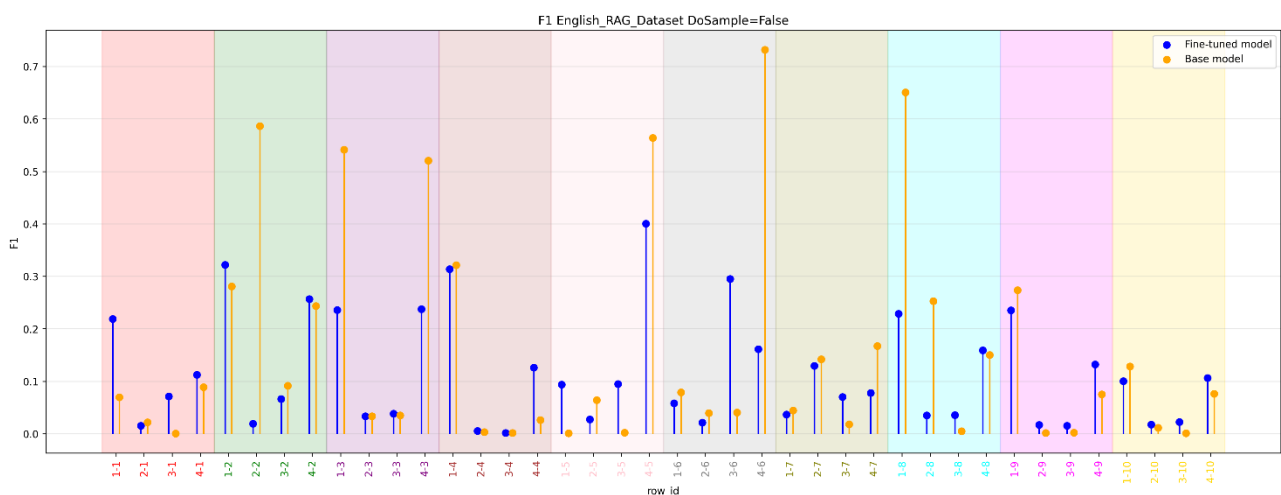
The mapping between **document_number** and **document** is provided below:

"Esperimento_1_Sienze_della_vita":1,
 "Esperimento_Elettricit ":2,
 "Esperimento_biologia_modificato":3,
 "Esperimento_elettromagnetismo_modificato":4,
 "Esperimento_fototropismo_modificato":5,
 "Esperimento_meccanica_modificato":6,
 "Estrazione_dna_vegetale_sintetico":7,
 "Manuale_energia_alternativa_modificato":8,
 "Manuale_la_termodinamica_modificato":9,
 "Saggio-fiamma_libro_zanichelli":10

The **question numbering** was assigned based on the order reported in the *English Questions* and *Italian Questions* sections.

Since each document contains 4 questions and there are 10 documents, each question within a document is assigned a value from 1 to 4.

For example, ‘Can you extract a list of the steps required to carry out Lavoisier’s law experiment and verify the principle of conservation of mass?’ is the first question of the first document; therefore, its question number is 1 and its document number is 1, resulting in a row ID of 1-1.



Example F1-Score Plot for the best configuration

row-id: question-document index

Example of how to read the plot:

Row-id: 1-1

Question: Can you extract a list of the steps required to carry out Lavoisier's law experiment and verify the principle of conservation of mass?'

Document: Esperimento_1_Scienze_della_vita.pdf

Row-id: 2-1

Question: What is the objective of the Lavoisier's law experiment regarding the conservation of mass in the reaction between zinc and hydrochloric acid?

Document: Esperimento_1_Scienze_della_vita.pdf

Row-id: 3-1

Question: I carried out the experiment leaving the flask open during the reaction between zinc crystals and hydrochloric acid: is this correct? If not, what should I correct to perform the experiment properly?

Document: Esperimento_1_Scienze_della_vita.pdf

Row-id: 4-1

Question: How can I correctly complete the experiment using a flask, zinc, and hydrochloric acid to verify that the total mass remains constant according to Lavoisier's law?

Document: Esperimento_1_Scienze_della_vita.pdf

Average Metrics of Configurations

Dataset	DoSample	Model	MeanPrecision	MeanRecall	MeanF1
English	False	Fine-tuned	0.3156	0.1091	0.1162
English	False	Base	0.4399	0.1549	0.1598
English	True	Fine-tuned	0.3212	0.1296	0.1291
English	True	Base	0.3828	0.1185	0.1302
Italian	False	Fine-tuned	0.2431	0.0458	0.0599
Italian	False	Base	0.2480	0.0744	0.0887
Italian	True	Fine-tuned	0.2113	0.0467	0.0601

Italian	True	Base	0.2549	0.1149	0.1120
English (DeepTranslator)	False	Base	0.2118	0.0669	0.0711
English (DeepTranslator)	False	Fine-tuned	0.2251	0.1023	0.1165
English (DeepTranslator)	True	Base	0.2351	0.0492	0.0571
English (DeepTranslator)	True	Fine-tuned	0.2285	0.1058	0.1155

Ranking of Configurations

Rank	File	Average ROUGE-S F1
1	ragEN_results_rouge_sDoSampleFALSE_ModelloBase.csv	0.1598
2	ragEN_results_rouge_sDoSampleTRUE_ModelloBase.csv	0.1302
3	ragEN_results_rouge_sDoSampleTRUE.csv	0.1291
4	rag_results_rouge_sDoSampleFALSE.csv	0.1165
5	ragEN_results_rouge_sDoSampleFALSE.csv	0.1162
6	rag_results_rouge_sDoSampleTRUE.csv	0.1155
7	rag_results_rouge_s_IT_DoSampleTRUE_ModelloBase.csv	0.1120
8	rag_results_rouge_s_IT_DoSampleFALSE_ModelloBase.csv	0.0887
9	rag_results_rouge_sDoSampleFALSE_ModelloBase.csv	0.0711
10	rag_results_rouge_s_IT_DoSampleTRUE.csv	0.0601
11	rag_results_rouge_s_IT_DoSampleFALSE.csv	0.0599
12	rag_results_rouge_sDoSampleTRUE_ModelloBase.csv	0.0571

Best Answer per Configuration

File	ROUGE-S F1 max	Question
rag_results_rouge_s_IT_DoSampleTRUE_ModelloBase.csv	0.8245	Qual è l'obiettivo dell'esperimento sui magneti riguardo all'interazione tra poli magnetici?

rag_results_rouge_s_IT_ DoSampleFALSE_ModelloBase.csv	0.8166	Come posso utilizzare un calibro a corsoio per misurare con precisione le dimensioni di un oggetto?
ragEN_results_rouge_s DoSampleFALSE_ModelloBase.csv	0.7317	How can I correctly complete the experiment using a vernier caliper to measure dimensions with precision?
ragEN_results_rouge_s DoSampleTRUE_ModelloBase.csv	0.6695	Can you extract the steps required to perform a combustion experiment inside a cylinder?
ragEN_results_rouge_s DoSampleTRUE.csv	0.5428	How can I verify Lavoisier's law using zinc and hydrochloric acid?
rag_results_rouge_s DoSampleFALSE_ModelloBase.csv	0.4641	Can you extract the steps required to perform a combustion experiment inside a cylinder?
ragEN_results_rouge_s DoSampleFALSE.csv	0.4002	How can I correctly complete the phototropism experiment using seeds and controlled light?
rag_results_rouge_s DoSampleFALSE.csv	0.4002	How can I correctly complete the phototropism experiment using seeds and controlled light?
rag_results_rouge_s DoSampleTRUE_ModelloBase.csv	0.3635	How can I perform a combustion experiment using alcohol and a spark source?
rag_results_rouge_sDoSampleTRUE.csv	0.3378	What is the objective of the distillation experiment?
rag_results_rouge_s_IT_DoSampleFALSE.csv	0.2328	Come posso eseguire correttamente un esperimento di combustione con alcol?
rag_results_rouge_s_IT_DoSampleTRUE.csv	0.1807	Come verificare la legge di Lavoisier con zinco e acido cloridrico?

Base vs Fine-Tuned Model Comparison

Model	Average ROUGE-S F1	Avg. Answer Length
Base	0.135	~968 characters
Fine-tuned	0.105	~1050 characters

Interpretation

The base model achieves higher ROUGE-S scores, while the fine-tuned model produces longer responses.

This suggests that:

- the base model tends to generate answers more closely aligned with the reference text
- the fine-tuned model tends to produce more elaborated and potentially more abstractive responses

However, the higher ROUGE scores of the base model are likely due to stronger lexical overlap rather than superior semantic understanding.

Limitations of ROUGE-S

ROUGE-S:

- measures lexical overlap
- does not capture semantic similarity
- penalizes paraphrased responses

Impact on IT vs EN Comparison

- English answers tend to have higher lexical overlap
- Italian answers are more variable and paraphrased

Therefore:

part of the advantage of the English dataset may stem not only from better retrieval, but also from higher compatibility with a lexical-based evaluation metric.

Impact on Model Comparison

- the base model tends to copy context, then higher ROUGE
- the fine-tuned model tends to rephrase, then lower ROUGE

This leads to a potential bias:

models producing more semantically rich responses may be penalized.

The combined quantitative and qualitative analysis shows that:

- retrieval is the most influential component
- errors in retrieval propagate directly to the generation phase
- the base model is favoured by ROUGE-S
- the fine-tuned model produces more elaborated responses
- the metric introduces bias toward lexical similarity

Moreover:

improving retrieval quality may yield larger performance gains than further optimizing the generative model.

Examples of Model Outputs

To better understand the differences between the base and fine-tuned models, this section presents selected examples of generated answers. These examples highlight key behaviors that are not fully captured by ROUGE-S scores.

Example 1: Procedural Task (Combustion Experiment)

Question:

How can the combustion experiment with alcohol and a spark be performed?

Fine-tuned model:

- Place the plexiglass cylinder
- Spray alcohol inside
- Close the cylinder
- Insert the ignition source
- Generate sparks

Base model:

- Insert the fuel into the cylinder
- Insert the spark source
- (repeated multiple times)

Observation:

The fine-tuned model correctly extracts and structures the procedure, while the base model

fails to generate a meaningful answer. This confirms the improvement in procedural reasoning and context grounding.

Example 2: Retrieval Usage (Magnet Experiment)

Question:

What are the steps to observe magnetic pole interaction?

Fine-tuned model:

- Take two magnets
- Bring the poles together
- Rotate one magnet
- Observe attraction and repulsion

Base model:

Not available in the provided context.

Observation:

Despite the information being present in the retrieved documents, the base model fails to use it. The fine-tuned model demonstrates improved ability to leverage the provided context.

Example 3: ROUGE Limitation (Plant Cells)

Question:

What is the difference between onion and orchid cells?

Fine-tuned model:

Onion cells are elongated, while orchid cells are involved in photosynthesis.

Base model:

Onion cells are elongated and flattened, while orchid cells are polygonal and contain chloroplasts.

Observation:

Both answers are correct, but the base model achieves a higher ROUGE score due to closer lexical similarity to the source text. This example highlights how ROUGE penalizes valid but more abstract responses.

Example 4: Loss of Detail (Distillation Experiment)

Question:

Describe how to build and use a distillator.

Fine-tuned model:

Build a system with containers and tubes, heat the mixture and collect the condensed liquid.

Base model:

Cut a latex glove, insert a tube, connect beakers, heat the mixture and observe condensation.

Observation:

The base model retains more technical details, while the fine-tuned model produces a more general but less precise description. This illustrates a trade-off between abstraction and technical accuracy.

LLM-as-a-Judge: Qualitative Evaluation Framework

To complement automatic evaluation metrics and overcome their limitations, a qualitative evaluation approach based on the LLM-as-a-Judge paradigm was adopted.

In this framework, a large language model (ChatGPT) is used as an evaluator to assess the quality of the generated answers. Unlike traditional metrics, which rely on surface-level similarity or embedding-based comparisons, this approach enables a more holistic evaluation of the responses.

Used Prompt:

"Ciao chat, ho il mio progetto che consiste nel: - Fare il fine tuning di un LLM (nel mio caso LLama 3.1 8B) su un dataset di 50000 documenti scientifici (chimica, biologia e fisica) ; - Fare la RAG con un corpus di 10 documenti scientifici (in italiano e in inglese + altre configurazioni del modello) e 4 domande per documento (per un totale di 40 domande); - Le risposte ottenute dalla RAG voglio farle valutare a te per capire se ogni risposta data dal mio LLM è coerente alla domanda, rispetto alle 40 domande totali; - Tieni conto che tutte le risposte sono in questo file csv, dove tra i vari campi si ha il campo 'document' che è il riferimento a questi documenti pdf che ti allego, che ho usato per la RAG. Valuta la risposta data dal mio modello, vedendo se è coerente con i documenti usati per la RAG."

Evaluation Methodology

Each generated answer was compared against the retrieved context and classified into one of the following categories:

- **Coherent**

The answer is fully aligned with the context, correctly captures the relevant information, and provides a complete and accurate response.

- **Partially coherent**

The answer is generally relevant but incomplete, imprecise, or missing key details from the context.

- **Not coherent**

The answer is incorrect, misleading, or not supported by the provided context.

This classification allows for a structured qualitative assessment that goes beyond lexical similarity.

Evaluation Criteria

The LLM-based evaluation implicitly considers multiple dimensions:

- **Semantic correctness**

Whether the answer accurately reflects the meaning of the context

- **Completeness**

Whether all relevant information is included

- **Context adherence**

Whether the response is grounded in the retrieved content

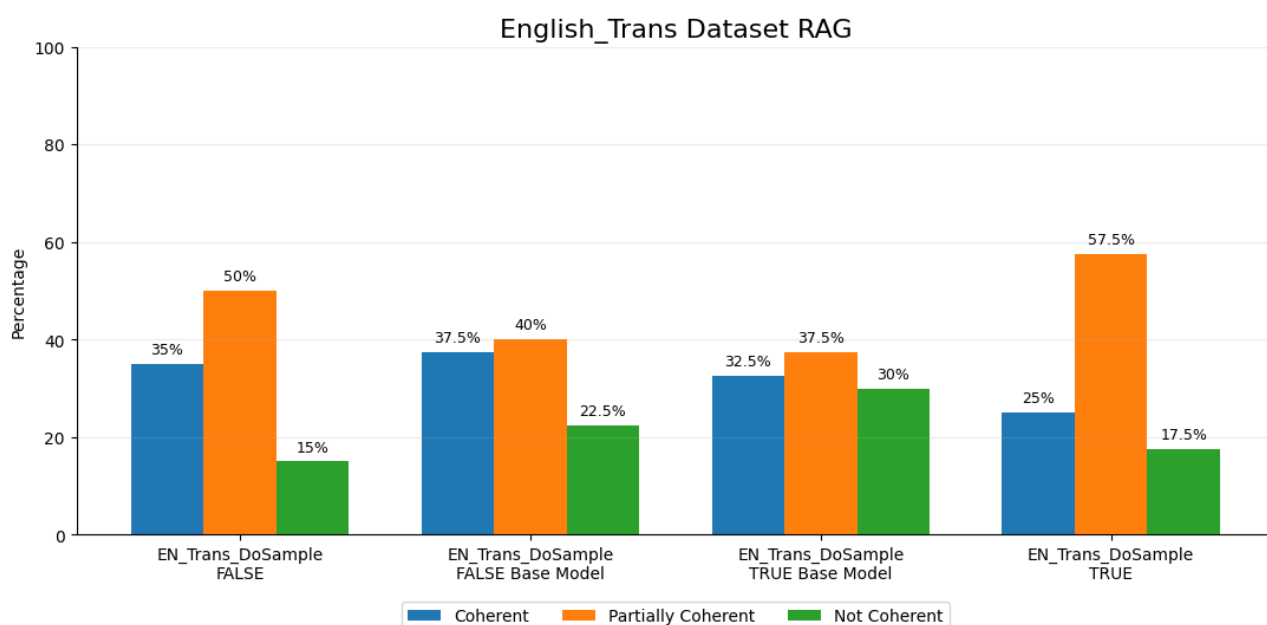
- **Clarity and usefulness**

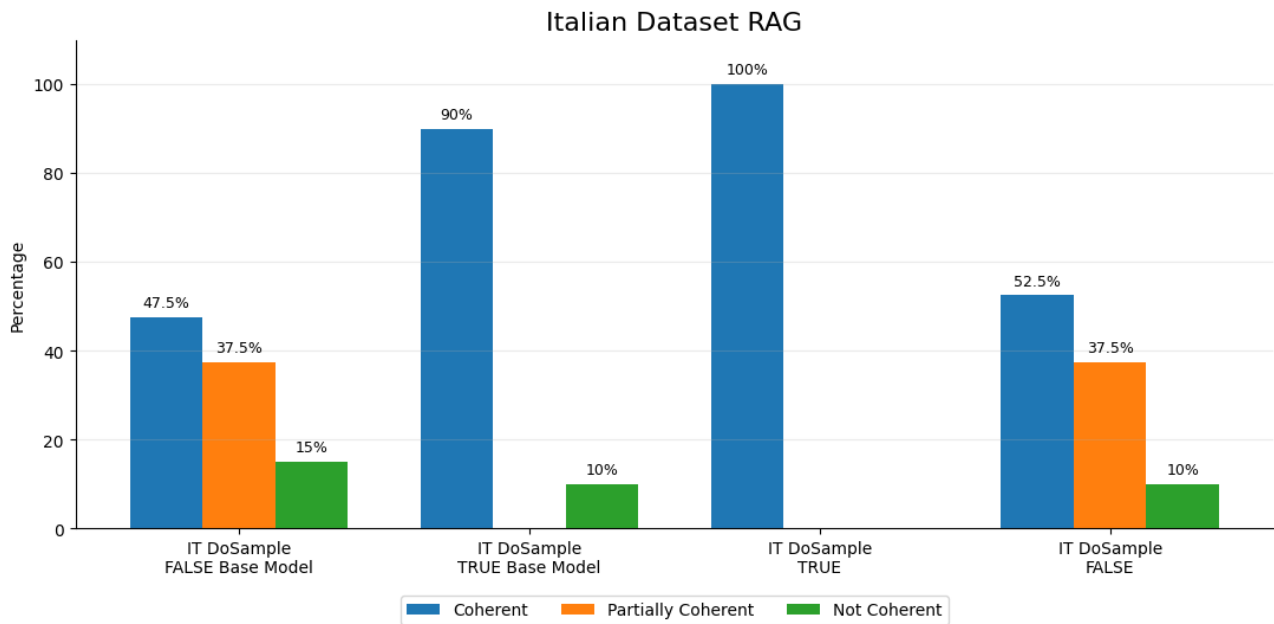
Whether the answer is understandable and practically informative

Italian/English (DeepTranslator) Dataset Rag

Experiment	% Coherent	% Partially Coherent	% Not Coherent	Description
EN_Trans_DoSample FALSE	35%	50%	15%	Mixed behavior: answers are sometimes correct but often incomplete or lacking precision
IT DoSample FALSE Base Model	47.5%	37.5%	15%	Mixed behavior, with moderate coherence but frequent lack of detail
IT DoSample TRUE Base Model	90%	0%	10%	Generally correct answers, well aligned with the context and

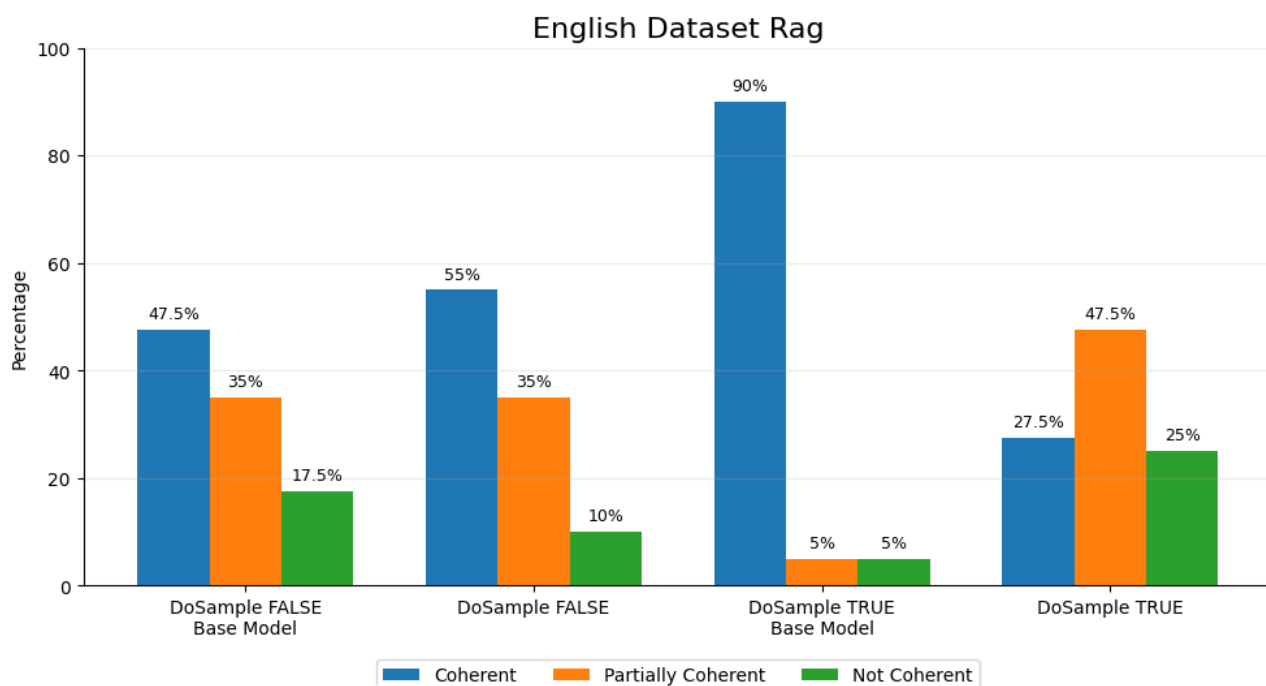
				with good completeness
IT DoSample TRUE	100%	0%	0%	Answers are always correct, complete, and perfectly aligned with the documents
EN_Trans_DoSample FALSE Base Model	37.5%	40%	22.5%	Mixed behavior with many partially coherent answers and some incorrect ones
EN_Trans_DoSample TRUE Base Model	32.5%	37.5%	30%	Weak performance: many incomplete answers and several errors
IT DoSample FALSE	52.5%	37.5%	10%	Good overall coherence, but still presence of partially complete answers
EN_Trans_DoSample TRUE	25%	57.5%	17.5%	Many partially coherent answers: incomplete retrieval of information from the context





English Dataset Rag

Experiment	% Coherent	% Partially Coherent	% Not Coherent	Description
DoSample FALSE Base Model	47.5%	35%	17.5%	Answers are generally correct and well aligned with the context, with good completeness
DoSample FALSE	55%	35%	10%	Answers are generally correct and well aligned with the context, with good completeness
DoSample TRUE Base Model	90%	5%	5%	Answers are generally correct and well aligned with the context, with good completeness
DoSample TRUE	27.5%	47.5%	25%	Answers are generally correct but often incomplete or insufficiently developed



Observed Benefits

The qualitative evaluation revealed patterns that were not captured by automatic metrics:

- Fine-tuned models often produce more coherent and structured answers, despite lower ROUGE scores
- Sampling strategies improve completeness and expressiveness
- Some answers with low lexical overlap are still fully correct and useful

These findings confirm the importance of including qualitative evaluation in the assessment of RAG systems.

BertScore

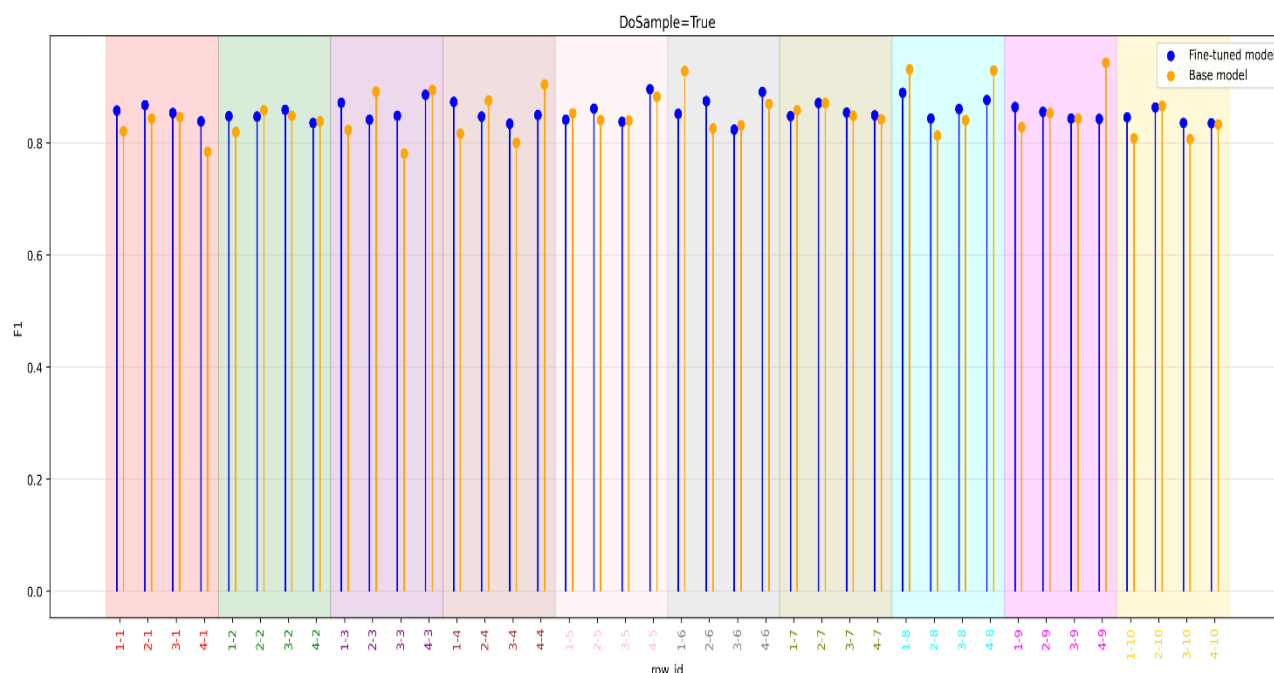
Parallel experiments were conducted using the BERTScore metric, which enables the evaluation of semantic similarity between the generated text and the reference, overcoming the limitations of metrics based solely on lexical overlap. The obtained results show an average score of approximately 0.8, indicating a generally good overall quality of the generated responses.

Additionally, a variation of the prompt was explored by making it less constrained and more open. This approach aims to encourage more natural and less rigid generation, allowing the model to produce responses that are potentially more articulated and better adapted to the context.

Ranking Based on BertScore

Dataset	Prompt	DoSample	Model	Mean Precision	Mean Recall	MeanF1
Italian	Base	True	Fine-tuned	0.887638	0.826327	0.855647
Italian	Modified	False	Fine-tuned	0.885508	0.821239	0.851852
Italian	Base	True	Base	0.880189	0.824467	0.851091
Italian	Base	False	Fine-tuned	0.883788	0.820333	0.850553
Italian	Modified	True	Fine-tuned	0.877045	0.82317	0.848942
Italian	Modified	False	Base	0.871848	0.820764	0.845168
Italian	Base	False	Base	0.875388	0.816746	0.84468
English	Modified	False	Base	0.88071	0.800077	0.837849
Italian	Modified	True	Base	0.861948	0.811776	0.83564
English	Base	False	Base	0.882273	0.794433	0.83541
English	Modified	True	Fine-tuned	0.87115	0.800938	0.834126
English	Base	True	Fine-tuned	0.872453	0.795506	0.831743
English	Base	False	Fine-tuned	0.87038	0.79386	0.829904
English	Modified	False	Fine-tuned	0.871806	0.790966	0.82897
English (DeepTranslator)	Base	False	Fine-tuned	0.859798	0.798849	0.827907
English	Modified	True	Base	0.872573	0.788627	0.827652
English (DeepTranslator)	Modified	False	Fine-tuned	0.85691	0.798654	0.826421
English (DeepTranslator)	Base	True	Fine-tuned	0.859867	0.795462	0.826076
English (DeepTranslator)	Modified	True	Fine-tuned	0.855272	0.797466	0.825074
English (DeepTranslator)	Modified	True	Base	0.848493	0.79484	0.819777
English (DeepTranslator)	Base	True	Base	0.855528	0.777962	0.81429
English (DeepTranslator)	Modified	False	Base	0.840096	0.784214	0.810234
English (DeepTranslator)	Base	False	Base	0.83723	0.779101	0.806416

English	Base	True	Base	0.772969	0.68504	0.725908
---------	------	------	------	----------	---------	----------



F1-Score Plot for the best experiment in the ranking.

N.B. on the x-axis question number are reported with the same logic of the ROUGE-S plot.

row-id: question-document index

Example of how to read the plot:

Row-id: 1-1

Question: Can you extract a list of the steps required to carry out Lavoisier's law experiment and verify the principle of conservation of mass?'

Document: Esperimento_1_Sienze_della_vita.pdf

Row-id: 2-1

Question: What is the objective of the Lavoisier's law experiment regarding the conservation of mass in the reaction between zinc and hydrochloric acid?

Document: Esperimento_1_Sienze_della_vita.pdf

Row-id: 3-1

Question: I carried out the experiment leaving the flask open during the reaction between zinc crystals and hydrochloric acid: is this correct? If not, what should I correct to perform the experiment properly?

Document: Esperimento_1_Sienze_della_vita.pdf

Row-id: 4-1

Question: How can I correctly complete the experiment using a flask, zinc, and hydrochloric acid to verify that the total mass remains constant according to Lavoisier's law?

Document: Esperimento_1_Sienze_della_vita.pdf

Limitations

Despite the comprehensive experimental setup, this study presents several limitations that must be acknowledged.

A primary limitation concerns the reliance on ROUGE-S as the main quantitative evaluation metric. While useful for measuring surface-level similarity, ROUGE-S fails to capture key aspects of response quality, including semantic correctness, completeness, and explanatory clarity. As a result, the metric introduces a systematic bias favouring models that reproduce text with high lexical overlap, potentially underestimating the performance of more abstractive approaches.

Although this limitation has been partially mitigated through the introduction of BERTScore and qualitative evaluation, the absence of human evaluation remains a notable constraint. Human judgment would provide a more reliable and nuanced assessment of the generated responses, particularly in educational contexts where clarity and correctness are essential.

Another limitation relates to the multilingual nature of the system. Even with the use of translation techniques, linguistic differences introduce variability that affects both retrieval and evaluation. Lexical-based metrics disproportionately penalize languages with higher variability and paraphrasing tendencies, such as Italian.

Finally, the dataset itself presents inherent constraints. Despite extensive filtering and validation, residual noise cannot be eliminated. Additionally, the imbalance across domains, especially the underrepresentation of Biology, may influence the generalizability of the results.

Conclusions

This study provides a comprehensive analysis of a Retrieval-Augmented Generation system, highlighting the interplay between retrieval quality, model adaptation, decoding strategies, and evaluation methodologies.

The findings demonstrate that conclusions drawn from lexical-based metrics alone can be misleading. While the base model appears to outperform fine-tuned configurations according to ROUGE-S, this advantage is largely attributable to superficial lexical similarity rather than genuine improvements in understanding or reasoning.

When incorporating semantic evaluation through BERTScore and qualitative assessment, a different trend becomes evident. Fine-tuned models, particularly when combined with stochastic decoding, exhibit superior performance in terms of coherence, completeness, and contextual grounding.

Moreover, the analysis reveals that original-language datasets, specifically the Italian dataset, yield higher-quality responses, despite being disadvantaged by lexical evaluation metrics. This highlights the importance of aligning evaluation methodologies with the true objectives of the system.

Considering these findings, this work underscores the necessity of adopting multi-dimensional evaluation frameworks that integrate lexical, semantic, and qualitative perspectives. Such an approach is essential for accurately assessing the capabilities of modern generative systems and for guiding future improvements in both model design and evaluation strategies.